Stelios Zacharioudakis

ML SYSTEMS ENGINEER | LEARNING AGENTS & APPLIED AI

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Machine learning engineer focused on reinforcement learning, self-improving systems, world models and LLM agents. Combine PyTorch research and verifiable evaluation with production software ownership, multilingual NLP and GPU inference engineering.

PROFESSIONAL EXPERIENCE

Paphos Medical Association - Head Engineer (Pro Bono) | Jun 2022 - Jul 2026

- Led architecture, development and delivery of AsklepiosMed, a donated production platform for member services and association administration.
- Integrated React interfaces, Node.js/Express APIs and PostgreSQL persistence; maintained access controls, administrative workflows and Linux deployment.

SELECTED PROJECTS AND RESEARCH

ForgeRL / ForgeBench - Reinforcement learning and agent evaluation

- Built a coding-agent evaluation platform with an offline fitted-Q routing controller, isolated execution and durable budget controls; compared six policies in a 162-episode transfer pilot and a separate 1,296-attempt follow-up.

AsklepiosMed - Multilingual NLP workflows

- Engineered multilingual NLP workflows for announcement drafting, email preparation, proofreading and text improvement within the association publishing platform.
- Implemented Greek-to-English, Russian and Simplified Chinese translation with terminology checks, structured output validation, fallback behavior and token-usage controls.

Inference Systems Lab - GPU serving evaluation

- Designed and ran 16,384 vLLM requests across RTX 4090 and A100 GPUs, two Qwen2.5 model sizes, cache modes and four repeats; measured throughput and latency, audited output parity and published reproducible aggregates.

MRI reconstruction - PyTorch and generative models | BSc thesis: Excellent, 10/10

- Reimplemented a published score-based reconstruction method with Flax-to-PyTorch checkpoint conversion; reproduced reference PSNR within 0.15 dB on 256 BraTS slices at 4x, 8x and 24x undersampling.
- Evaluated uncertainty and retrospective domain shift; released reproducible evidence and independent CPU ensemble diagnostics. Original pretrained prior credited; no clinical validation.

MTA-Scan - Streaming data and temporal evaluation

- Built live feed ingestion, a FastAPI/Mapbox workspace and bounded historical storage; automated chronological 15/30-minute proxy forecasting with baselines and freshness gates. November audit: 44 daily exports and 28 evaluated days.

TECHNICAL SKILLS

ML & AI: Python, PyTorch, reinforcement learning, LLM agents, world models, generative models, NLP, vLLM, NumPy, SciPy, scikit-learn and experimental evaluation.

Engineering: FastAPI, React, Node.js/Express, JavaScript, PostgreSQL, Redis, Linux, Nginx, Docker, Git and automated testing.

EDUCATION AND ONGOING RESEARCH

BSc in Computer Science | National and Kapodistrian University of Athens, Jun 2026

Degree: Very Good | Thesis: Excellent (10/10), deposited September 2026.

Independent research: verifier-guided self-improvement, continual learning for coding agents, latent world models and policy evaluation; ongoing and unpublished.